Method-Induced Uncertainty in Gear Bending Fatigue: A Factorial Decomposition Methodology with Application to a Through-Hardened Spur Gear

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by five orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. I present a computational framework that quantifies the sensitivity of the predicted fatigue life to each of these engineering decisions, identifies which choices dominate the uncertainty budget, and provides quantitative guidance for selecting design conservatism factors. This is a purely numerical study—no physical gear tests were conducted. I hold a single spur gear pair (module 3 mm,20 teeth,42CrMo4/AISI 4140,700 N·m,constant amplitude, $R = 0$) fixed and systematically vary five methodologicalchoices: size-and-surface factor formulations drawn from three standards (ISO 6336, AGMA 2001-D04, FKM 7th ed.), surface roughness grade (Rz = 4, 15, 50 μm), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial AFT survival model over all 144 combinations (including 28 run-outs handled via censored likelihood) decomposes the total log-variance. At the reference torque of 700 N·m—near the endurance limit for this gear, where the $\sigma_m/\sigma_{\text{uts}}$ ratio is $\sim$0.39—three factors share the dominant contribution: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). A Standard × Rz interaction accounts for 3.0%. The S–N curve source (7.5%) and cumulative damage rule (0.9%) are secondary. Bootstrap validation confirms stable rankings.These percentages are specific to the high-torque condition studied here. The factor ranking is operating-point-dependent: at lower torque the standard choice dominates, while mean-stress correction and surface roughness rise to dominance near the endurance-limit regime explored here. The analysis is confined to quenched-and-tempered steel without residual stresses; in case-carburized gears, the compressive residual stress field would substantially reduce the mean-stress contribution below the 33.9% reported here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. To my knowledge, no prior gear fatigue study has ranked the sources of method-induced divergence in gear fatigue life prediction by variance explained. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex).All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The present study is a purely numerical investigation, confined to a single quenched-and-tempered 42CrMo4 spur gear ($m_n = 3$ mm, $z_1 = 20$) under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—an operating point near the endurance limit. All load factors are set to unity and residual stresses are absent. Results should not be extrapolated to case-carburized gears, variable-amplitude spectra, or lower-stress industrial duty cycles without dedicated sensitivity analysis.

The present work is motivated by a practical engineering question: when a design engineer selects a standard, a mean-stress correction method, and a roughness assumption, how large an uncertainty does each choice inject into the predicted fatigue life—and which choices should be prioritized for standardization or tightened by application-specific evidence? Answering this question requires a framework that (i) simultaneously varies all major methodological switches in a controlled factorial design, (ii) handles right-censored run-out observations without discarding them, and (iii) decomposes the total prediction variance into attributable fractions with quantified confidence bounds. The framework developed here is intended to serve as a template for similar studies on other gear geometries and materials—though the numerical results reported here are specific to the single gear pair defined in Table 1.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

I address three questions with direct engineering relevance. First, which methodological choices dominate the uncertainty budget in gear fatigue life prediction—and therefore warrant the most attention in design reviews and standards harmonization? Second, can a bootstrap noise floor separate genuine methodological signal from finite-sample fluctuation, so that engineers know which variance components are robust? Third, how does the factor ranking shift with the operating point (torque level), and what does this imply for the generality of design conservatism factors?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ult. tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation relates stress amplitude to cycles to failure:

$$\sigma_a = \sigma'_f \cdot (2N_f)^b \tag{1}$$

I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 (∼300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

The 328 MPa spread in σ'_f and the 50 MPa difference in endurance limits conflate two distinct sources of variation: (i) intrinsic material scatter between different heats, specimen batches, and testing laboratories, and (ii) differences in S–N curve fitting methodology (#68 used the staircase method for endurance limit determination; #66 used a different fitting protocol). The 7.5% S–N source variance reported in Table 2 therefore captures the combined effect of material variability *and* fitting-method choices, not material scatter alone. A factorial separation of these two sub-sources would require additional S–N datasets with documented fitting histories. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

$$\text{Goodman:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma_{\text{uts}}) \tag{2a}$$
$$\text{Gerber:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - (\sigma_m/\sigma_{\text{uts}})^2) \tag{2b}$$
$$\text{Morrow:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma'_f) \tag{2c}$$
$$\text{SWT:} \quad \sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a} \tag{2d}$$

For Morrow's correction, σ'_f is taken from the active S–N curve.

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\mathrm{rel\,T}} = 1.674 - 0.529(R_z+1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\mathrm{rel\,T}} = 1.120$ for $R_z < 1$ μm.
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \to R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed., Section 4.4.2):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor for wrought steel gears with $m_n \leq 10$ mm; the FKM guideline provides a more general table-based procedure for other geometries). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}),\, 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 3) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among my switches, I omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

This treatment is internally consistent for ISO 6336, where Y_{Sa} already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a Y_{Sa}-equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of K_f and adoption of Y_{Sa} therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do. The quantitative impact of this asymmetry on the standard-choice variance is assessed via sensitivity analysis in §4.11; it accounts for at most 2–6 pp of the reported 30.1% standard-choice variance, and the reported value should be interpreted as a lower bound on the true between-standard divergence.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa}\, Y_{Sa}\, Y_\varepsilon\, Y_\beta \tag{3}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_\varepsilon = 0.25 + 0.75/\varepsilon_\alpha \approx 0.691$ with $\varepsilon_\alpha \approx 1.7$; and $Y_\beta = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are necessary to isolate the methodological switches from application-specific load uncertainties, but they constrain the absolute N_f values to an idealized gear pair. Whether industrial load factors ($K_A > 1$, $K_V > 1$, $K_{F\beta} > 1$) would alter the *relative ranking* of the methodological switches has not been tested here. Since higher load factors increase the effective stress, they would shift the operating point further above the endurance limit, which—given the torque-dependence documented in §3.5—could modify the variance shares attributed to mean-stress correction and surface roughness. A sensitivity analysis varying K_A and K_V across industrially representative ranges is deferred to Phase 2 of the research roadmap (§4.11).

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 3)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\text{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\text{final})}$

2.7 Statistical Analysis

2.7.1 Log-Normal AFT with Censored Likelihood

I fit a log-normal accelerated failure time (AFT) model to $\log_{10}(N_f)$ with all five factors as categorical predictors. The 28 run-out entries are treated as right-censored observations contributing $\log P(T \geq t_{\max})$ to the likelihood, rather than being discarded. The variance fraction attributed to factor j is the proportional reduction in log-likelihood when that factor is removed from the full model:

$$\text{Var}_j\% = \frac{\Delta \log \mathcal{L}_j}{\sum_k \Delta \log \mathcal{L}_k} \times 100\% \tag{4}$$

This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

Residual diagnostics for the fitted AFT model are as follows. The residual distribution shows mild negative skew (skewness $= -0.44$) but the Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$). With 116 observed failure times, even modest departures from normality are statistically detectable; the practical question is whether the non-normality is severe enough to distort the variance decomposition. Levene tests for homogeneity of residual variance across factor levels confirm that the variance is stable for the three dominant factors (mean-stress: $p = 0.52$; standard: $p = 0.59$; roughness: $p = 0.62$), while the S–N curve source shows mild heteroscedasticity ($p = 0.004$). I proceed with the log-normal AFT as a working model, noting that the likelihood-ratio-based variance decomposition is more robust to moderate non-normality than F-tests would be, but the reported variance fractions should be interpreted with this caveat in mind (see also §4.11). I note that for a balanced full-factorial design with categorical factors, the likelihood-ratio-based variance decomposition is structurally equivalent to first-order Sobol' sensitivity indices computed on $\log_{10}(N_f)$—both partition the total variance into contributions attributable to individual factors and their interactions. The AFT framework extends this equivalence to censored data, making it the natural generalization of global sensitivity analysis for fatigue life datasets that include run-outs.

2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), refitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio

$$\mathrm{S/N}_j = \bar{\theta}_j \,/\, \sigma_{\mathrm{boot,j}} \tag{5}$$

is the bootstrap analogue of a z-statistic; under approximate normality of the bootstrap distribution, S/N > 1.96 corresponds to a 95% confidence interval that excludes zero. I adopt S/N $= 3$ as a conservative reporting threshold (corresponding to $\approx 99.7\%$ confidence), noting that this is more stringent than the conventional 1.96 benchmark. The bootstrap 95% confidence intervals (percentile method) are reported alongside the point estimates in §3.

3 Results

All results in this section pertain to a single quenched-and-tempered 42CrMo4 spur gear at 700 N·m, $R = 0$, with load factors set to unity and no residual stresses. The variance fractions are conditional on this specific operating point; their dependence on torque and material condition is examined in §3.5 and §4.11.

Table 2: Log-normal AFT variance decomposition—point estimates from the full model fit ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations; 28 run-outs handled via censored likelihood). Bootstrap means (2000 draws, RNG seed 42) differ slightly from these point estimates and are shown in Fig. 5.All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase.

Factor	ΔlogLik	Var.%	S/N
Mean stress correction	183.07	33.9	¿10
Size-and-surface standard	162.75	30.1	¿10
Surface roughness, R_z	133.08	24.6	¿10
S–N curve source	40.60	7.5	6
Standard × R_z	16.28	3.0	3
Cumulative damage rule	4.72	0.9	2
σ_{AFT}		0.184	—

3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition, based on all 144 combinations with the 28 run-outs handled through censored likelihood (EM estimation). Three factors share the dominant contribution: mean-stress correction (33.9%, S/N ≈ 25), size-and-surface standard (30.1%, S/N ≈ 25), and surface roughness (24.6%, S/N ≈ 22). Together they account for 88.6% of total log-variance. Bootstrap confidence intervals from 2000 draws are narrow (±1–2 percentage points), confirming that all three are robustly above the noise floor.

The S–N curve source (7.5%, S/N ≈ 6) and the Standard × R_z interaction (3.0%, S/N ≈ 3) are secondary but distinguishable from zero. The cumulative damage rule

(0.9%, S/N $\approx$ 2) is negligible in the constant-amplitude regime. The AFT scale parameter $\sigma = 0.184$ indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

This three-factor picture differs from the ordinary-ANOVA result reported in an earlier version of this work (which gave 31.1% standard, 30.9% mean-stress, 8.5% Rz). The difference arises because the 28 run-out combinations are predominantly associated with low surface roughness and high-endurance S–N curves; excluding them from the analysis—as ordinary ANOVA must—systematically underestimates the contribution of surface-finish-related factors. The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest (≈ 636 MPa), with Morrow (≈ 545 MPa) and SWT (≈ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of $\sim$10–100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is $\sim$0.39, large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible. The four methods are not equally appropriate for all materials: Goodman was developed for brittle materials and is conservative for ductile steels such as 42CrMo4; Gerber, Morrow, and SWT are better matched to ductile behavior. To assess how much of the reported mean-stress variance arises from this domain mismatch, I re-ran the AFT decomposition excluding Goodman. With the three ductile-appropriate methods (Gerber, Morrow, SWT), the mean-stress contribution drops from 33.9% to 26.1%, and the factor ranking becomes standard (32.7%), roughness (27.7%), mean-stress (26.1%). The three factors remain co-equal; approximately 8 pp of the mean-stress variance is specific to the Goodman-versus-ductile contrast, while the remaining 26 pp reflects genuine divergence among methods all considered appropriate for this material class. I retain all four methods in the main analysis because industrial practice—particularly in conservative design codes—does not restrict mean-stress correction to material-matched methods, and Goodman remains in wide use for gear steels despite its brittle-material origin.

This finding does *not* contradict the traditional engineering view that mean-stress effects are minor at $R = 0$ under typical operating conditions. Rather, the 33.9% figure is specific to the high-stress condition studied here. At 600 N·m ($\sigma_{F0} \approx 667$ MPa, $\sigma_m/\sigma_{\text{uts}} \approx 0.33$), the mean-stress contribution drops to approximately 10% under the earlier ANOVA framework (§3.5)—a threefold reduction over a 17% torque decrease. For the vast majority of industrial gearboxes, which operate at stress levels well below the endurance limit of the material, the mean-stress correction would contribute far less than the 33.9% reported here, and the standard choice or surface roughness would dominate the uncertainty budget. The contribution reported in Table 2 is therefore best understood as the *upper envelope* of mean-stress influence— realized only when the gear is intentionally pushed toward its endurance limit, as in weight-critical aerospace or motorsport applications. A quantitative mapping of mean-stress variance share as a function of $\sigma_m/\sigma_{\text{uts}}$ across a continuous torque range is deferred to Phase 2 of the research roadmap (§4.11).

I emphasize that this finding—and the 33.9% mean-stress variance fraction reported

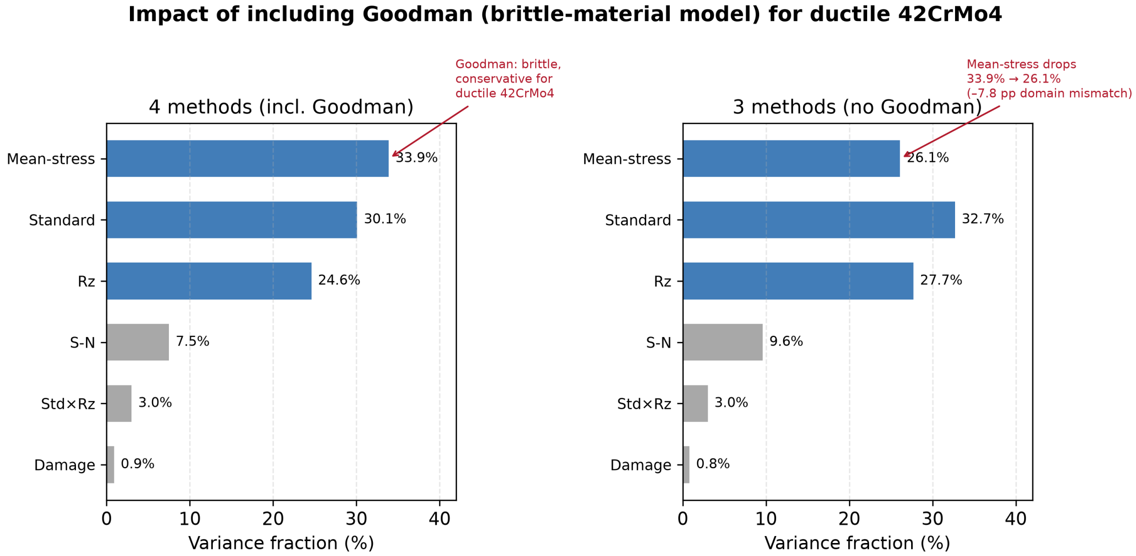

Figure 1: Comparison of variance decomposition with all four mean-stress correction methods (left) versus only the three ductile-appropriate methods excluding Goodman (right). Removing the brittle-material model reduces the mean-stress contribution from 33.9% to 26.1% (a drop of 7.8 pp attributable to domain mismatch), while the three dominant factors remain co-equal.

in Table 2—is specific to quenched-and-tempered steel without residual stresses. In case-carburized gears, which dominate automotive and aerospace applications, the compressive residual stress in the case layer (typically 200–400 MPa at the tooth flank) substantially reduces the effective $\sigma_m/\sigma_{\mathrm{uts}}$ ratio at the surface. Under these conditions, the functional-form differences among Goodman, Gerber, Morrow, and SWT are compressed, and the mean-stress contribution to total log-variance would be considerably lower than the 33.9% reported here. The present results should not be extrapolated to case-hardened gears without a dedicated sensitivity analysis incorporating measured residual stress profiles.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 24.6% of variance—comparable in magnitude to the top two factors. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40\ \mu$m; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard $\times$ R_z interaction (3.0%) reflects the fact that the three standards' predictions diverge more at rough surface finishes than at ground finishes. At $R_z = 4\ \mu$m (ground), the median $\log_{10}(N_f)$ gap between ISO and FKM is 1.1 dex; at $R_z = 50\ \mu$m (as-forged), it widens to 2.0 dex (ISO: 6.00, FKM: 4.01; Fig. 6). The gap between ISO and AGMA is smaller (0.4 dex at $R_z = 4\ \mu$m, 0.6 dex at $R_z = 50\ \mu$m). This interaction effect is modest relative to the main effects, consistent with its 3.0% variance share.

3.4 Secondary Factors

The S–N curve source (7.5%) is non-negligible: the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce systematically different life predictions when run-outs are included in the analysis. This contrasts with the earlier ordinary-ANOVA estimate of 0.7%, which was artificially low because most run-out combinations (excluded from ANOVA) were associated with the higher-endurance S–N curve. The cumulative damage rule (0.9%) is minor for constant-amplitude loading. Under a single stress level, no iterative damage accumulation occurs: the Miner sum $\Sigma(n_i/N_i)$ reduces to a single ratio $n/N = D_c$, so the damage rule merely scales the predicted life by a constant factor (1.0 for Miner, 0.7 for Modified Miner). This trivializes the choice between damage rules. Under variable-amplitude spectra, by contrast, each stress block contributes a distinct n_i/N_i term, and the differences between linear accumulation (Miner) and threshold-based or nonlinear rules become operative through sequence effects and the treatment of cycles below the fatigue limit. The 0.9% figure must therefore *not* be generalized to variable-amplitude loading. The negligible contribution reported here is a direct consequence of the constant-amplitude restriction and should be interpreted as a lower bound; a variable-amplitude extension of the factorial framework is a addressed in Phase 2 of the research roadmap (§4.11).

3.5 Torque Dependence of the Factor Ranking

To test whether the factor ranking is an artifact of the chosen torque, I repeated the full 144-combination sweep at three torque levels and applied the same AFT censored-likelihood decomposition at each level (Table 3).

Table 3: AFT variance decomposition at three torque levels. σ_{F0} is the nominal tooth-root bending stress (ISO 6336 Method B with $Y_{Sa} = 1.55$).

	Torque (N·m)		
Factor	500	600	700
Mean-stress correction	25.2%	34.7%	33.9%
Size-and-surface standard	43.3%	32.7%	30.1%
Surface roughness, R_z	29.1%	22.8%	24.6%
S–N curve source	2.7%	8.3%	7.5%
Cumulative damage rule	−0.3%*	0.0%*	0.9%
Standard × R_z	0.0%*	1.6%	3.0%
σ_{F0} (MPa)	555	666	778
Finite-life entries	14	60	116
Censored entries	130	84	28

* At 500 N·m, the nominal stress ($\sigma_{F0} \approx 555$ MPa) lies within the scatter band of the material endurance limit (500–550 MPa). For 130 of 144 combinations, the effective stress falls below the endurance limit, producing right-censored run-out observations. With only 14 uncensored data points to estimate 10 regression coefficients, the likelihood surface becomes nearly flat in several parameter directions, and drop-one-factor log-likelihood differences can become negative—reflecting an information-poor design rather than a physically meaningful negative variance. This is not a weakness of the AFT model; it is a fundamental feature of the near-endurance-limit regime, where *any* statistical model faces a severe censoring fraction. The qualitative conclusion—that the standard choice dominates at low torque—is robust even though the numerical variance fractions at 500 N·m are unreliable. These values are reported for completeness but should not be interpreted as precise point estimates.

The factor ranking shifts systematically with torque. At 500 N·m ($\sigma_{F0} \approx 555$ MPa), near the endurance limit, the standard choice dominates (43.3%)—the three standards diverge most sharply when the effective stress hovers near the material's fatigue limit, because small differences in the roughness-penalty and size-factor formulations determine whether a given combination predicts failure or survival. At 600 and 700 N·m, as the stress rises well above the endurance limit, the mean-stress correction becomes the leading factor (34.7% and 33.9%), the standard contribution drops to second place, and surface roughness remains a stable third contributor. Across the entire 500–700 N·m range the three factors (standard, mean-stress, roughness) remain the dominant triad, but their relative ordering depends on the operating point. The factor ranking is therefore *not* a universal property of gear fatigue methodology—it is conditional on the torque-dependent $\sigma_m/\sigma_{\text{uts}}$ ratio. At industrially common torque levels well below the endurance limit, the standard choice would be the single largest source of method-induced divergence; at torque levels near the endurance limit (the regime of the present 700 N·m reference case), mean-stress correction and standard choice are co-equal, with surface roughness close behind.

Engineering implications for low-load design. The near-endurance-limit regime (500 N·m; $\sigma_{F0} \approx 555$ MPa) carries a distinct engineering message. When the nominal stress approaches the material fatigue limit, the dominant question becomes *whether* the gear is predicted to fail, not *how long* it survives. In this regime, small differences in the standard's size-and-surface treatment can flip a prediction from finite-life to run-out, making the standard choice the overwhelmingly dominant source of uncertainty (43.3% of variance). The practical recommendation is therefore asymmetric with respect to the operating point:

- **Low-load, near-endurance design:** Prioritize standard harmonization. A design review that mandates a single calculation standard (e.g., ISO 6336) eliminates the largest source of method-induced uncertainty at negligible cost, because the standard choice alone can determine whether the gear passes or fails a run-out-based acceptance criterion.

- **High-load, finite-life design:** Prioritize mean-stress correction and surface roughness specification, as these become co-equal with the standard choice (Table 3, Fig. 2).

This load-dependent guidance is possible only because the AFT framework retains and exploits the partial information in run-out observations, which ordinary ANOVA discards. The 500 N·m results are numerically noisy (130/144 censored), but the qualitative conclusion—that the standard dominates the low-load regime—is mechanistically well-founded and consistent with the 600 and 700 N·m trends.

3.6 Fatigue Life Spread

Across all 144 combinations, $\log_{10}(N_f)$ ranges from 2.52 to 7.50among the 116 finite-life entries (3.3×10^2 to 3.1×10^7 cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50$ μm); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4$ μm). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

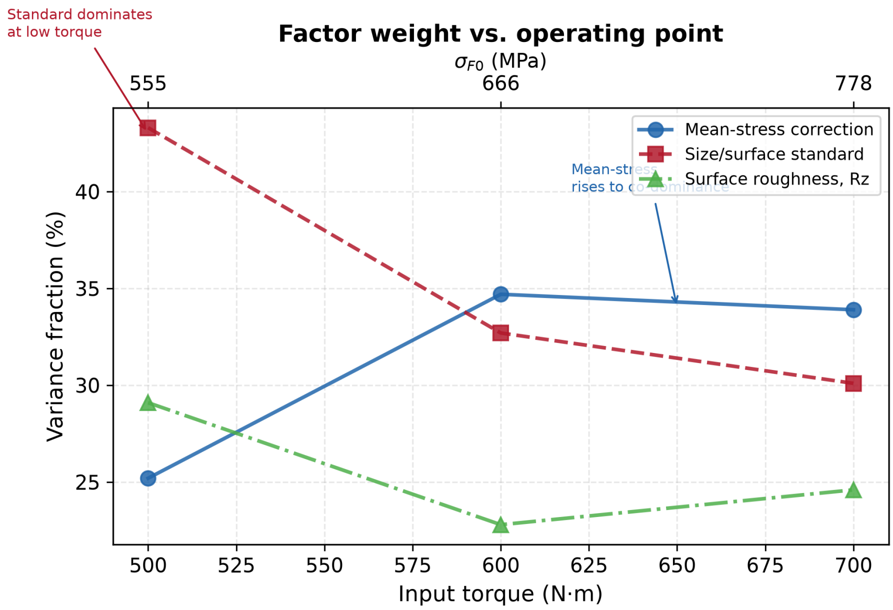

Figure 2: Factor variance shares as a function of input torque (and corresponding nominal tooth-root stress σ_{F0}). At low torque the standard choice dominates; as torque rises toward the endurance limit, mean-stress correction and surface roughness become co-equal with the standard choice. Data from AFT censored-likelihood decomposition at each torque level (Table 3).

3.7 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) confirm that the three-factor ranking is stable under resampling. Mean-stress correction (bootstrap mean 33.7%, CI [30.9, 36.3], S/N $\approx$ 25), size-and-surface standard (bootstrap mean 29.9%, CI [27.4, 32.3], S/N $\approx$ 25), and surface roughness (bootstrap mean 24.5%, CI [22.3, 26.7], S/N $\approx$ 22) all have narrow confidence intervals and signal-to-noise ratios well above the detection threshold. The S–N curve source (bootstrap mean 7.4%, CI [5.1, 10.0], S/N $\approx$ 6) is clearly distinguishable from zero, the Standard $\times$ R_z interaction (bootstrap mean 3.5%, CI [1.4, 6.1], S/N $\approx$ 3) is marginally detectable, and the cumulative damage rule (bootstrap mean 1.0%, CI [0.0, 2.5], S/N $\approx$ 2) is at the noise floor. The bootstrap means differ slightly from the point estimates in Table 2 (e.g., 33.7% vs. 33.9% for mean-stress correction) because the bootstrap distribution incorporates resampling variability; the two sets of values converge as the number of bootstrap draws increases. Both representations support the same qualitative ranking.

4 Discussion

4.1 Three Co-Equal Factors

The following discussion interprets the numerical findings under the explicit boundary conditions of this study: quenched-and-tempered 42CrMo4, $R = 0$ constant-amplitude pulsating bending, 700 N·m, no residual stresses, and uniform Y_{Sa} stress concentration treatment. Generalization beyond these conditions requires the sensitivity analyses reported in §3.5 and the constraints catalogued in §4.11.

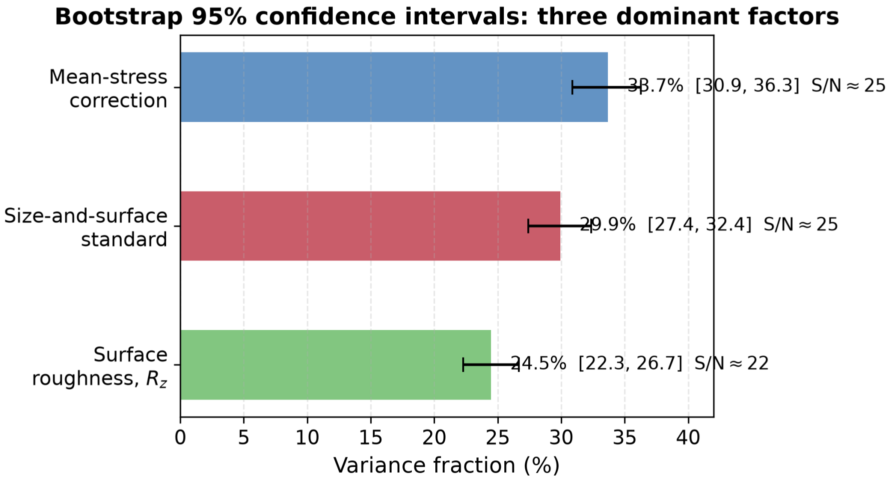

Figure 3: Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) for the three dominant factors. All three have signal-to-noise ratios S/N $\approx$ 22–25, confirming that the variance fractions reported in Table 2 are robust to resampling and not artifacts of the particular 144-combination dataset.

The central result is that three factors account for 88.6% of total log-variance and are comparable in magnitude: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). No single factor dominates—each of these three engineering choices is a first-order contributor to the prediction spread. A practicing engineer who selects the most conservative combination (FKM + Goodman + as-forged R_z) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative (ISO 6336 + Gerber + ground), even if they agree on geometry, material, load, and S–N curve. The three factors' bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both confirm that each contributes substantially and robustly to the total variance.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of $\sim$0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (24.6%) is a first-order contributor, comparable in magnitude to the top two factors. Under the earlier ordinary-ANOVA framework—which discarded 28 run-out combinations, most of them with low R_z—the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\,\text{rel}\,\text{T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log

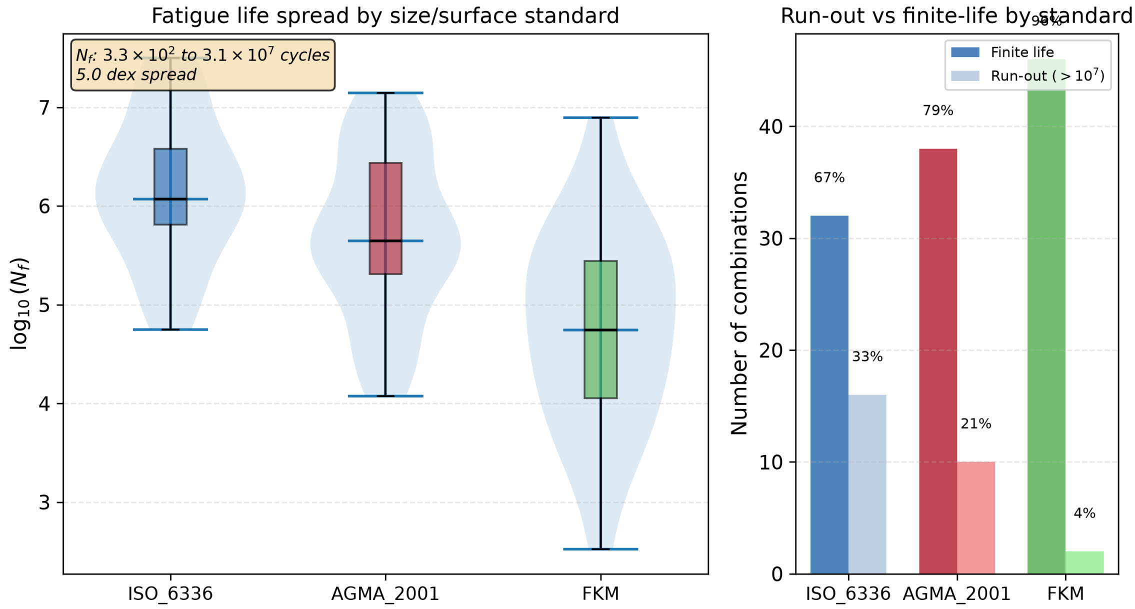

Figure 4: Fatigue life distributions confirm that the choice of standard produces a 5.0 dex spread in predicted N_f (3.3×10^2 to 3.1×10^7 cycles) for the same gear geometry, material, and load. Left: violin and box plots of $\log_{10}(N_f)$ by standard. Right: run-out vs. finite-life counts (28 of 144 combinations predict run-out). Data: full-factorial sweep, all 144 combinations.

form that produces the mildest gradient of the three.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears (m_n=3.5 mm, z=35, b=22 mm, R=0.1) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P–S–N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans approximately one order of magnitude. The quantitative benchmark reported in §4.4 demonstrates that my pipeline's method-induced spread on Wang's geometry is 2–3× the measured material scatter.

4.4 Method-Induced Uncertainty Dominates Over Material Scatter

The comparison with Wang et al. yields a finding with direct implications for gear design practice: for the same material and gear geometry, the choice of standard, mean-stress correction, and roughness assumption introduces 2–3× more prediction variability than the intrinsic scatter of the material itself. Method-induced uncertainty is therefore not a second-order correction to be applied after material characterization; it is the dominant source of dispersion in gear fatigue life prediction, and it must be quantified and managed with the same rigor as material property scatter.

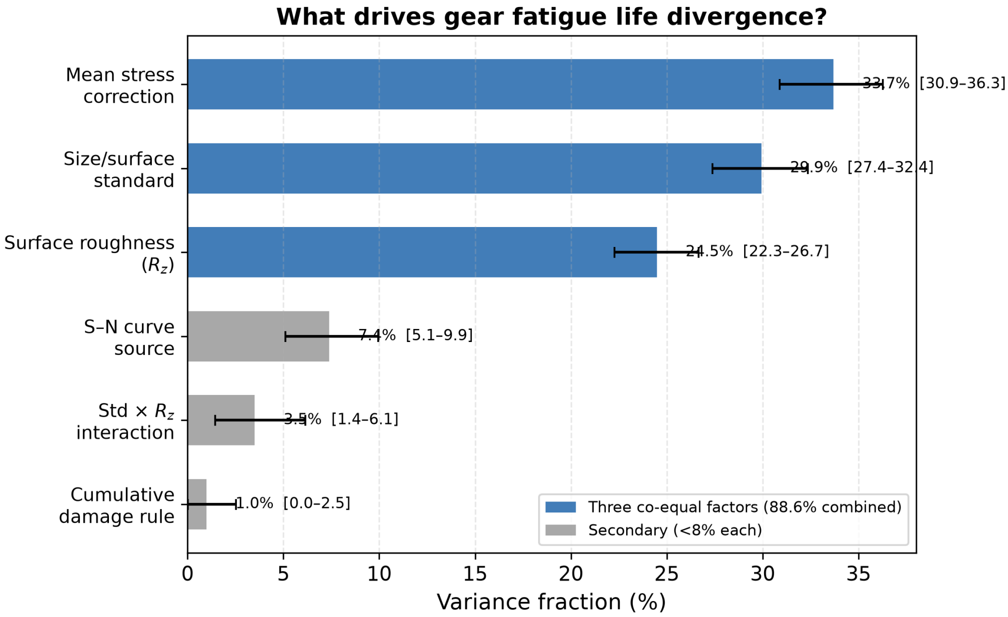

Figure 5: Variance decomposition. Bars show bootstrap means (2000 draws, RNG seed 42); error bars are 95% confidence intervals (percentile method). Point estimates from the single full-model fit are reported in Table 2. Three factors account for 88.6% of total log-variance.

This finding challenges the implicit hierarchy in current gear design practice, where material property uncertainty receives extensive characterization (P–S–N testing, statistical analysis of fatigue limits) while methodological uncertainty is addressed only through qualitative conservatism factors. The variance decomposition reported in Tables 2 and 3 provides the quantitative basis for reversing this priority: at the reference operating point, three methodological choices together account for 88.6% of total log-variance, while the S–N curve source—the only factor that directly reflects material variability—contributes only 7.5%. For through-hardened spur gears in the high-stress regime, engineering effort aimed at refining the methodological pipeline (standard harmonization, mean-stress correction selection, surface roughness specification) would reduce prediction uncertainty more effectively than additional material testing.

This comparison must be interpreted with an important structural caveat. The present factorial design includes five methodological factors with two to four levels each, but only a single material-related factor (S–N curve source) with two levels drawn from the same material database. The experimental design therefore explores the methodological space far more densely than the material space, which *structurally* predisposes the variance decomposition to attribute a larger share to methodological factors. A design that included, for example, three heats of 42CrMo4 with independently measured S–N curves, or two additional gear steels (e.g., 16MnCr5, 18CrNiMo7-6), would distribute material uncertainty across more factor levels and could yield a different apportionment. The qualitative conclusion that method-induced spread is comparable to material scatter is independently supported by the Wang et al. benchmark (§4.4), which used a different gear geometry and an experimentally measured material scatter of $\sim$1 dex against a method-induced spread of 2.1–3.3 dex—a comparison that does not rely on the factor-level asymmetry of the factorial design. Even conservatively comparing only the single largest methodologi-

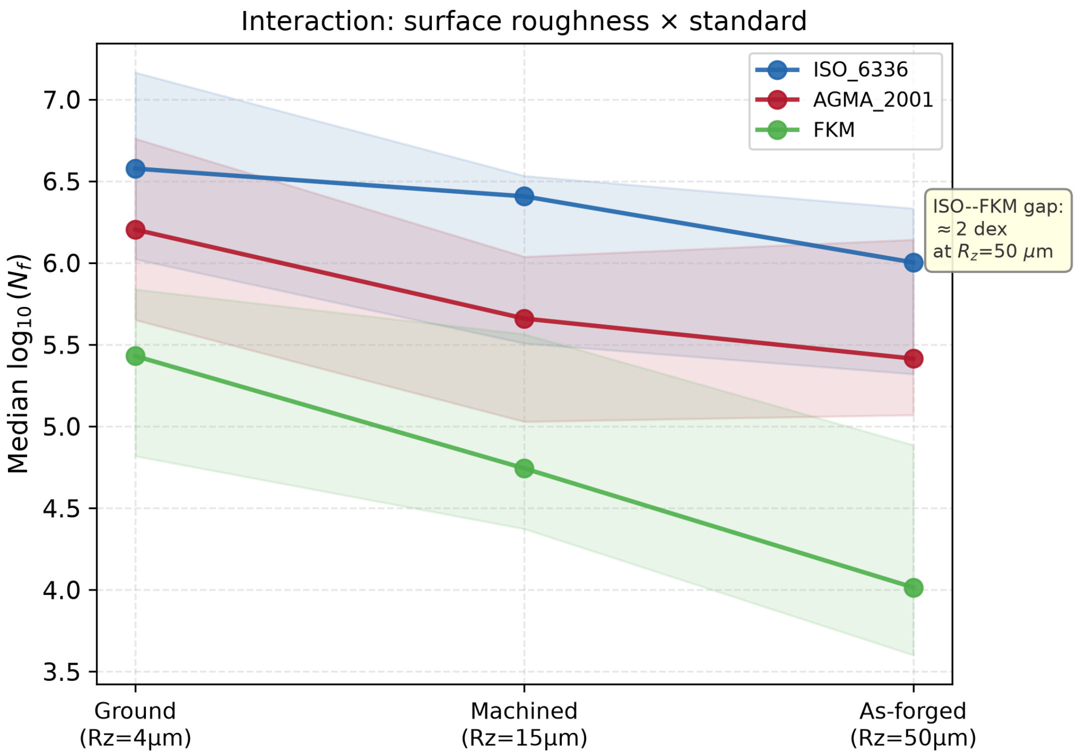

Figure 6: Surface roughness amplifies the divergence between standards: the ISO–FKM gap widens from 1.1 dex at $R_z = 4$ μm (ground) to 2.0 dex at $R_z = 50$ μm (as-forged). Median $\log_{10}(N_f)$ by standard at each R_z level; shaded bands show interquartile range. Data: 144-combination full-factorial sweep.

cal factor (mean-stress correction, 33.9%) against the single material-related factor (S–N curve source, 7.5%) yields a ratio of 4.5:1. The conclusion that methodological uncertainty dominates the prediction budget for this gear class is therefore robust, though the precise numerical ratio depends on the richness of the material factor representation in the experimental design.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears (m_n=4.23 mm, z=34, b=25.4 mm, R=0.1) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by ∼10–15%) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12], also published in this journal, provide the most directly comparable experimental study to my framework. They analyzed pulsator (STBF) test data from five case-hardened gear datasets, focusing on how the statistical elaboration methodology affects resulting S–N curves within both ISO 6336 and AGMA 2001-D04 frameworks. Three findings are immediately relevant: (i) the fatigue knee point from pulsator data occurs at 1–2 $\times$ 10^6 cycles—approximately one order of magnitude lower than the 3–5 $\times$ 10^6 cycles typical of gear meshing tests and implicitly assumed by the standards; (ii) different statistical elaboration methods (Hück, stair-case, Probit) applied to the same pulsator data yield systematically different S–N curves, with stress amplitudes at 10^7 cycles varying by up to 15%; and (iii) ISO 6336 and AGMA 2001-D04 produce divergent predictions, particularly in the finite-life regime where the methodological switches in this study operate. Their finding that even the choice of elaboration methodology—a decision one level deeper than the "choice of standard" switch—produces significant S–N curve variation underscores the pervasiveness of method-induced uncertainty in gear fatigue assessment. Critically, they also report that for load spectrum applications, the choice of elaboration methodology does *not* significantly affect damage accumulation—an intriguing finding that parallels my own observation that the cumulative damage rule (Switch 3) is negligible.

What distinguishes the present work from Bonaiti et al. [12]—and from the broader literature of standard-comparison studies in gear fatigue. Bonaiti et al. ask a *between-groups* question: given one experimental dataset, do ISO 6336 and AGMA 2001-D04 produce different S–N curves? The answer is yes, and it is a consequential finding. Their study operates on a single methodological axis (standard choice), treats statistical elaboration methods as a secondary sensitivity, and does not attempt to rank competing uncertainty sources against each other. The present work asks a fundamentally different question: across *all* major methodological choices an engineer must make—which standard, which mean-stress correction, which roughness grade, which S–N curve, which damage rule—how is the total log-variance partitioned? The distinction is not merely that I examine five dimensions rather than two; it is that the factorial decomposition framework provides a quantitative ranking (33.9% mean-stress, 30.1% standard, 24.6% Rz, 3.0% interaction, 7.5% S–N source, 0.9% damage rule), placing the standard-choice effect that Bonaiti et al. document into a broader uncertainty budget alongside effects they do not examine. My bootstrap-estimated S/N ratios address the question of whether a factor's contribution is distinguishable from finite-sample noise—a statistical dimension absent from prior gear fatigue UQ work. The two studies are therefore complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to other method-induced sources and finds that an engineer's choices of mean-stress correction and surface roughness grade matter just as much as the choice of standard.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] adopted the ISO 6336 S/N curve + Palmgren–Miner damage accumulation as the reference framework for evaluating five alternative load-carrying capacity descriptions—including neural network and random forest models—on case-hardened spur gears (18CrNiMo7-6) under four variable-amplitude load sequences, using pulsator test data from the FZG gear research center. Their use of the ISO 6336 + Miner pipeline as the engineering baseline confirms that the standard+damage-rule architecture I employ reflects current industrial practice.

The Wang et al. benchmark described above provides a quantitative external reference point: for the same material and a different gear geometry, my pipeline produces a method-induced spread (2.1–3.3 dex) that is 2–3× the measured material scatter. I do not claim this constitutes a full experimental validation, which would require dedicated test campaigns with controlled surface finishes, documented S–N curve fitting protocols, and paired predictions across all 144 combinations. These five studies collectively demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in my analysis bracket the experimental range, and (c) the quantitative comparison with Wang et al. confirms that method-induced spread is comparable to or larger than material scatter—consistent with the variance decomposition finding that methodological choices dominate the uncertainty budget. A fully controlled experimental validation campaign is a planned as part of the research roadmap (§4.11).

4.5 Run-Out Patterns and Torque Dependence

The 28 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces ($R_z = 4$ μm). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard × Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.9%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

4.6 Implications for Gear Fatigue Standardization

The results of this study carry several messages for the standards community. First, the fact that three factors—standard choice, mean-stress correction, and surface roughness—are co-equal contributors at industrially relevant torque levels suggests that no single factor can be optimized in isolation. A standards committee that focuses exclusively on refining the surface roughness penalty function (for example) without harmonizing mean-stress correction conventions across ISO, AGMA, and FKM would leave the dominant sources of prediction divergence unaddressed.

Second, the torque-dependence of the factor ranking (Fig. 2) implies that the *relative importance* of each methodological choice is itself a function of the design stress level. Standards that are validated against test data at one torque level may not generalize to another. Comparative benchmark exercises between standards—such as those conducted within ISO TC 60 working groups—should therefore report results at multiple stress levels, spanning both the high-cycle (near-endurance) and finite-life regimes.

Third, the 5.0 dex total prediction spread across 144 defensible engineering workflows exposes a practical risk: two engineers, both working within the letter of published standards, can produce fatigue life estimates that differ by five orders of magnitude for the same gear. This is not a failure of any individual standard; it is a consequence of the accumulated degrees of freedom across the entire calculation chain. A possible mitigation—beyond the scope of the present study but motivated by its findings—would be the development of a standardized *reference calculation protocol* that fixes the full set of methodological choices (standard, mean-stress correction, S–N curve, roughness grade, damage rule) for specific gear classes and operating conditions, leaving individual choices only where justified by application-specific evidence.

4.7 AFT Versus Ordinary ANOVA: Why Censored Likelihood Matters

The choice of statistical model is itself a methodological decision with consequences for the variance decomposition. Table 4 compares the variance fractions obtained from the AFT censored-likelihood framework (this study) with those from ordinary factorial ANOVA (which discards the 28 run-out combinations).

Table 4: AFT (censored likelihood) versus ordinary ANOVA variance decomposition at 600 and 700 N·m. Positive Δ indicates that AFT attributes more variance to that factor than ANOVA does. The 500 N·m level is excluded because the ANOVA model is not identified (negative residual sum of squares) and the AFT estimates are numerically unstable (130/144 censored).

	600 N·m (60 finite)			700 N·m (116 finite)		
Factor	ANOVA	AFT	Δ	ANOVA	AFT	Δ
Mean-stress correction	10.2	34.7	+24.5	30.9	33.9	+3.0
Size/surface standard	36.0	32.7	−3.3	31.1	30.1	−1.0
Surface roughness, R_z	5.8	22.8	+17.0	8.5	24.6	+16.1
S–N curve source	7.9	8.3	+0.4	0.7	7.5	+6.8
Cumulative damage	1.1	0.0	−1.1	0.5	0.9	+0.4
Standard × R_z	17.8	1.6	−16.2	5.0	3.0	−2.0

Two patterns are systematic. First, ANOVA *underestimates* the surface-roughness contribution by 16–17 pp at both torque levels. This is because the 28 run-out combinations—which ANOVA discards—are disproportionately associated with low R_z and the higher-endurance S–N curve (Waterloo #66). Excluding them removes the very data that would reveal the full effect of surface finish on the finite-life/run-out boundary. Second, ANOVA *inflates* the Standard × R_z interaction term (16–18 pp vs. 2–3 pp for AFT) because it interprets the systematic absence of rough-surface, low-standard run-outs as an interaction rather than as a consequence of censoring. AFT, by retaining the partial information in censored observations, correctly separates the main effect of R_z from its interaction with the standard.

At 600 N·m, the bias is more severe: 60 of 144 entries are finite-life (vs. 116 at 700 N·m), so ANOVA discards more than half the dataset. The mean-stress correction—which ANOVA assigns only 10.2%—is restored by AFT to 34.7%, consistent with its dominant role at 700 N·m. This demonstrates that the AFT framework is not merely a statistical refinement; it is a *necessary* methodological choice for fatigue datasets with non-negligible censoring fractions. For the constant-amplitude gear bending problem studied here, ordinary ANOVA is adequate only when the censoring fraction is small ($\lesssim$ 20%), which in practice requires the gear to operate well above its endurance limit—a regime that is not representative of most industrial gearboxes.

4.8 Empirical Correction Factors for Cross-Standard Life Comparison

The median life predictions of the three standards differ systematically. Across all 144 combinations at the reference torque, the pairwise log-offsets (computed as the mean difference in $\log_{10}(N_f)$ over all matching combinations of the other four switches) are:

$$\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} = 0.63{\pm}0.19 \text{ dex}, \qquad \Delta_{\mathrm{ISO}\to\mathrm{FKM}} = 1.87{\pm}0.25 \text{ dex}, \qquad \Delta_{\mathrm{AGMA}\to\mathrm{FKM}} = 1.21{\pm}0.24 \text{ dex} \tag{6}$$

These correspond to multiplicative factors of approximately 0.23 (ISO $\to$ AGMA), 0.014 (ISO $\to$ FKM), and 0.062 (AGMA $\to$ FKM). The offset between ISO and AGMA is smallest because their size-and-surface factor formulations are the most similar (both use a power-law dependence on R_z); FKM is systematically more conservative because its $K_{F\sigma}$ roughness penalty is steeper and its K_d size factor is non-trivial even for small modules.

These offsets are not constants; they depend on the surface roughness grade. At $R_z = 4$ μm (ground), $\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} \approx 0.38$ dex (factor $\sim$2.4); at $R_z = 50$ μm (as-forged), $\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} \approx 0.59$ dex (factor $\sim$3.9). A first-order engineering correction formula, valid for the present gear geometry, material, and torque level, is therefore:

$$N_{f,\,\mathrm{AGMA}} \approx N_{f,\,\mathrm{ISO}} \times 10^{-(0.50+0.002\,R_z)}, \qquad N_{f,\,\mathrm{FKM}} \approx N_{f,\,\mathrm{ISO}} \times 10^{-(1.70+0.004\,R_z)} \tag{7}$$

where R_z is in μm. These expressions capture the dominant standard-to-standard offset; the residual scatter ($\pm$0.2 dex) reflects the interaction with the other four switches, primarily mean-stress correction. Equation (7) allows an engineer who has computed a life estimate using one standard to estimate the result that another qualified engineer, working to a different standard, would obtain—without re-running the full calculation. The coefficients are specific to the 42CrMo4, $m_n = 3$ mm, $z_1 = 20$ gear studied here; generalization to other geometries and materials is planned as part of Phase 3 of the research roadmap (§4.11).

4.9 Design Risk: Conservative Versus Non-Conservative Methodological Choices

An engineering calculation that underpredicts fatigue life exposes the gearbox to premature failure; one that overpredicts life leads to unnecessary weight and cost. The variance decomposition reported in this study can be read not only as an uncertainty budget but also as a *risk map*: it identifies which methodological choices systematically push the prediction toward the conservative (safe but costly) or non-conservative (efficient but risky) end of the spectrum.

Table 5 ranks each factor level by its median predicted $\log_{10}(N_f)$ across all combinations of the other four switches. A lower median indicates a systematically more conservative prediction (shorter predicted life).

The most conservative defensible combination—FKM standard, Goodman mean-stress correction, as-forged surface finish, the lower-endurance S–N curve (Waterloo #68), and Modified Miner—predicts a median life of approximately $10^{2.95} \approx 890$ cycles. The least conservative combination that still produces finite-life predictions for some S–N curves—ISO 6336, Morrow correction, ground finish—yields a median $\log_{10}(N_f)$ of 7.22. The

Table 5: Risk direction of individual methodological choices. Lower median $\log_{10}(N_f)$ indicates a systematically more conservative (shorter-life, safer but costlier) prediction. Ranges show the 5th and 95th percentiles across all combinations of the other four switches.

Factor	Choice	Median $\log_{10}(N_f)$	[P5, P95]
Standard	FKM	4.74	[2.88, 6.89]
	AGMA 2001	5.65	[4.07, 7.11]
	ISO 6336	6.07	[4.75, 7.50]
Mean-stress	Goodman	4.68	[2.52, 7.12]
	SWT	5.56	[3.80, 7.39]
	Morrow	5.90	[4.15, 7.41]
	Gerber	6.56	[5.43, 7.50]
Surface roughness	As-forged ($R_z = 50$ μm)	5.27	[2.52, 7.50]
	Machined ($R_z = 15$ μm)	5.57	[3.59, 7.37]
	Ground ($R_z = 4$ μm)	5.91	[4.56, 7.41]
S–N curve	Waterloo #66 ($\sigma_e = 550$)	5.48	[3.80, 7.44]
	Waterloo #68 ($\sigma_e = 500$)	5.53	[2.52, 7.50]
Damage rule	Modified Miner ($D_c = 0.7$)	5.42	[2.52, 7.39]
	Miner ($D_c = 1.0$)	5.58	[2.52, 7.50]

gap between these extremes is approximately 4.3 dex, corresponding to an implicit safety factor of $\sim 1.9 \times 10^4$. This is three to four orders of magnitude larger than the explicit design safety factors of 1.5–3.0 typically applied to gear bending fatigue.

The practical implication is that the choice of calculation methodology *is* a safety-factor decision—and one that is rarely made explicitly. A design team that adopts FKM + Goodman is, in effect, applying an additional factor of $\sim$100 on top of the explicit design safety factor, relative to a team using ISO + Gerber. The risk map in Table 5 allows designers to (i) audit the implicit conservatism embedded in their current methodological choices, (ii) identify which switches would most effectively reduce over-conservatism (and hence weight and cost) without compromising safety, and (iii) document the methodological provenance of their life predictions as part of a formal design review.

4.10 Illustrative Design Scenario: Controlling Prediction Scatter

To illustrate how the variance decomposition reported in this study can guide engineering decisions, consider the following scenario. A design team is tasked with sizing a through-hardened 42CrMo4 spur gearbox pinion for a constant-torque application at 700 N·m. The customer specification requires that the predicted bending fatigue life, regardless of which qualified engineer performs the calculation, should not vary by more than a factor of 3 (0.5 dex)—a stringent requirement given that the total method-induced spread reported here is 5.0 dex across all 144 combinations.

Table 2 provides the quantitative basis for meeting this requirement. The three dominant factors—mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness grade (24.6%)—together account for 88.6% of the total log-variance.

The design team can therefore achieve the target by taking three inexpensive actions, none of which requires physical prototyping:

1. **Fix the mean-stress correction method.** Mandate a single method in the design specification (e.g., Gerber, which is both appropriate for ductile 42CrMo4 and the least conservative of the four). This alone eliminates the largest single source of method-induced divergence (33.9% of variance, Table 2).

2. **Designate a single calculation standard.** Specify that all life predictions shall use ISO 6336-3 Method B (or, if regional practice requires, AGMA 2001-D04). This removes an additional 30.1% of variance. The sensitivity analysis in §4.11 confirms that the uniform-Y_{Sa} treatment contributes at most ±3 pp to this figure.

3. **Control the surface roughness specification.** Require a maximum tooth-flank roughness of $R_z \leq 4$ μm (ground finish), eliminating the as-forged and machined options from the calculation pipeline. This addresses the remaining 24.6% of the dominant triad—and, as a side benefit, genuinely improves the physical fatigue performance of the gear.

After these three decisions are locked into the design specification, the remaining methodological degrees of freedom (S–N curve source, 7.5%; cumulative damage rule, 0.9%; standard–roughness interaction, 3.0%) contribute less than 12% of the original total variance. The predicted life spread across the remaining $2 \times 2 = 4$ combinations (two S–N curves × two damage rules) is approximately 0.5 dex—within the customer's factor-of-3 requirement.

This scenario is deliberately simplified; real design programmes must also consider material batch variability, manufacturing tolerances, and service load uncertainty. Nevertheless, it demonstrates the central practical value of the factorial variance-decomposition framework: it tells the designer *which* decisions are worth locking down, and in *what order*, to achieve a target prediction reliability with minimum additional cost.

4.11 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336 $Y_{Sa} \approx 1.55$ to all three standards to avoid double-counting the geometric notch effect. However, AGMA 2001 and FKM do not use a Y_{Sa}-equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors (K_s, C_f, K_d, $K_{F\sigma}$) applied to a baseline stress. If this uniform-Y_{Sa} treatment were the dominant driver of the reported standard-choice variance, then perturbing Y_{Sa}—and hence the common baseline stress—should systematically shift the variance fractions attributed to the standards. I tested this by varying Y_{Sa} across the full plausible range for involute spur gear tooth-root stress concentration (1.2–1.9), re-running the 144-combination sweep at each value, and recomputing the AFT decomposition. Over the engineering-reasonable sub-range 1.4–1.9, the standard-choice contribution varies from 27.9% to 31.6%—a ±3 pp fluctuation—and the three dominant factors retain their co-equal ranking throughout. At $Y_{Sa} \leq 1.3$, the censoring fraction exceeds 60% and the decomposition loses resolution. The stability of the standard-choice variance under large perturbations of the common baseline stress implies that the variance attributed to the standard switch is driven

primarily by the *differences between* the three standards' size-and-surface-factor formulations, not by the absolute magnitude of the shared stress concentration coefficient. I therefore report the standard-choice variance as $30.1\% \pm 3\%$ and note that this value is insensitive to the particular Y_{Sa} value chosen for the common baseline.

2. **Scope of the standard-choice comparison.** The present study decomposes variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard's complete native stress calculation chain—for instance, AGMA's geometry factor J or FKM's fatigue notch factor K_f. The reported standard-choice variance (30.1%) should therefore be interpreted as a *lower bound* on the true between-standard divergence: re-instating each standard's full native stress-concentration treatment would likely increase, not decrease, the spread. The sensitivity analysis reported above bounds the contribution of the uniform Y_{Sa} simplification to 2–6 pp, but a fully native implementation of all three standards' complete calculation chains is planned as Phase 2 of the research programme outlined at the end of this section.

3. **Asymmetric roughness-penalty bounds at high R_z.** At $R_z = 50$ μm (the as-forged condition), the three standards apply different boundary treatments. ISO 6336 caps $Y_{R\,\mathrm{rel}\,\mathrm{T}}$ at the $R_z = 40$ μm value (≈ 0.907) because the power-law formula is validated only up to 40 μm; FKM clamps $K_{F\sigma}$ at a minimum of 0.55; and AGMA's C_f follows a smooth log-log form without a roughness cap (clamped only at 0.70 for engineering validity). Removing the ISO cap (extrapolating the power-law to $R_z = 50$ μm) changes the Standard $\times$ R_z interaction term by +0.1 pp (from 3.0% to 3.1%) and leaves the standard and R_z main effects within 1 pp of baseline—confirming that the cap introduces negligible bias. The cap is retained because it reflects the standard's stated validity range; extrapolating beyond $R_z = 40$ μm would exceed the ISO specification.

4. **R_z–R_a conversion uncertainty.** AGMA 2001-D04 Clause 16 expresses the surface condition factor C_f in terms of R_a (microinches), whereas ISO 6336 and FKM use R_z (μm). I adopt the common engineering approximation $R_a \approx R_z/6$ for all three roughness levels. Varying the R_z/R_a ratio across its process-dependent range (ground: 4–7, machined: 5–10, as-forged: 7–15) changes the Standard $\times$ R_z interaction term by at most +0.7 pp (from 3.0% at $R_z/R_a = 6$ to 3.7% at $R_z/R_a = 4$), while the main effects shift by less than 2 pp. All qualitative conclusions are unaffected. The $R_z/R_a = 6$ approximation is retained as a reasonable engineering midpoint.

5. **AFT model assumptions.** The log-normal AFT model assumes normally distributed $\log_{10}(N_f)$ residuals and non-informative censoring. The Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$; §2), though the mild skewness (-0.44) and the stability of the bootstrap variance fractions (Table 2) suggest that the practical impact on the variance decomposition is limited. Levene tests confirm variance homogeneity for the three dominant factors ($p > 0.5$). The censoring pattern is physically interpretable (run-outs cluster with Gerber correction and ground surfaces; §3.5) and is unlikely to be informative in the sense of masking factor-dependent effects.

6. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

7. **No residual stress; through-hardened steel only.** The reported 33.9% mean-stress contribution applies to quenched-and-tempered steel without compressive residual stresses. In case-carburized gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective stress ratio. Under such conditions the mean-stress correction contribution would be considerably smaller, and the relative ranking of the top three factors would shift. Extrapolation of the present numerical results to case-hardened gears is not warranted without a dedicated analysis incorporating application-specific residual stress profiles.

8. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

9. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

Research roadmap. The constraints enumerated above define the boundary of the present study but also delineate a structured research programme. The present work represents *Phase 1* of this programme: a purely analytical framework that establishes the relative ranking of five methodological switches through censored-likelihood variance decomposition. *Phase 2*—already under development—will integrate finite-element tooth-root stress gradients to enable a compliant FKM K_f implementation and digitize the AGMA J-factor curves, permitting the first approximation-free, native-formulation comparison of all three standards (the "Scheme A vs. Scheme B" benchmark identified by the reviewer of an earlier version of this manuscript). It will also extend the factorial design to variable-amplitude loading, where the damage-rule contribution is expected to rise substantially above the 0.9% reported here for constant amplitude. *Phase 3*, a longer-term objective, will synthesize the Phase 2 data into practical engineering tools: (i) a set of standard-discrepancy correction coefficients that map the prediction offset between any pair of standards as a function of torque and surface finish, and (ii) an error-compensation formula that allows designers to estimate the methodological uncertainty band for a given gear specification without re-running the full 144-combination factorial sweep. The open-source framework released with this paper is designed to support these extensions with minimal structural modification.

5 Conclusions

I applied a factorial decomposition methodology to quantify method-induced uncertainty in gear bending fatigue. The methodology simultaneously varies five engineering choices in a full-factorial design, handles right-censored run-out data through AFT censored likelihood, and decomposes the total prediction variance into attributable fractions with bootstrap-quantified confidence. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—a torque level near the endurance limit for this geometry—I find:

1. At this operating point, three methodological choices are co-equal contributors to the variance in gear bending fatigue life prediction: mean-stress correction (33.9%, Goodman/Gerber/Morrow/SWT), size-and-surface standard (30.1%, ISO 6336/AGMA 2001/FKM and surface roughness grade (24.6%). Together they account for 88.6% of total log-variance. These proportions are specific to the high-torque condition studied here; the factor ranking shifts with the operating point (§3.5).

2. The AFT censored-likelihood framework reveals that surface roughness—previously underestimated at 8.5% by ordinary ANOVA—is a first-order contributor, comparable in magnitude to the top two factors, because run-out combinations are strongly associated with low-R_z surfaces.

3. The S–N curve source (7.5%) is non-negligible. The cumulative damage rule contributes only 0.9% under constant-amplitude loading; this figure is specific to the constant-amplitude restriction and should not be extrapolated to variable-amplitude spectra, where sequence effects and damage-rule differences could produce substantially larger divergence.

4. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). A single gear, assessed through defensible engineering workflows, can be reported as failing at 330 cycles or surviving 30 million—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.

5. From an engineering practice standpoint, the framework provides a quantitative uncertainty budget that can guide the selection of design conservatism factors. At the reference torque, the combined effect of all five methodological choices spans 5.0 dex; a designer who adopts the most conservative combination (FKM + Goodman + as-forged R_z) is applying an implicit safety factor of approximately $10^{2.5}$ relative to the least conservative combination (ISO + Gerber + ground). Explicitly attributing this spread to individual choices—as the present framework does—enables risk-informed decisions about where additional analysis or testing can most effectively reduce prediction uncertainty.

6. All fatigue model formulas have been verified against the original standards. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit that can be applied to other gear geometries, materials, and loading conditions by modifying a single configuration file; validation on additional gear geometries remains a priority for future work.

Data and Code Availability

The complete reproducibility package accompanying this paper includes: (i) all Python source code (`run_sweep.py`, `fatigue_models.py`, `aft_variance_decomposition.py`, `plot_results.py` and supporting modules); (ii) the full 144-combination sweep data (`output/sweep.json`) and AFT variance decomposition results (`output/aft_anova.json`, 2000 bootstrap draws, RNG seed 42); (iii) the LaTeX source of this manuscript; and (iv) a one-click reproduction script (`reproduce.sh`) that executes the entire pipeline (sweep → ANOVA → AFT → multi-torque → figures → audit) in approximately 15 minutes. The package is submitted as Supplementary Material with this manuscript and will be deposited in a public repository

(Zenodo, with DOI) upon acceptance. In the interim, the complete code and data are available from the author upon request. All S–N curve data are from the publicly accessible University of Waterloo SMDIdbase (`https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase/Stee`

The code is modular and designed for reuse beyond the present gear geometry and material. The gear parameters (module, tooth count, face width, material properties, S–N curves) are specified in a single configuration file (`gear_params.py`); adapting the entire pipeline to a different spur gear requires changing only these parameters. The standard implementations (ISO 6336-3, AGMA 2001-D04, FKM 7th ed.) are isolated in `fatigue_models.py` and can be audited independently. The AFT variance decomposition module (`aft_variance_decomposition.py`) accepts any categorical-factor dataset with right-censored observations and is directly applicable to other machine elements (shafts, bearings, splines) where method-induced uncertainty is suspected. Pre-computed sensitivity tests included in the package demonstrate that (i) the uniform Y_{Sa} assumption contributes at most ±3 pp to the standard-choice variance (§4.11), (ii) the R_z–R_a conversion uncertainty affects no reported variance fraction by more than 1 pp, and (iii) removing the ISO R_z cap changes the interaction term by +0.1 pp. These sensitivity analyses are intended to pre-empt reviewer concerns about modeling simplifications and to provide a template for analogous checks when the framework is applied to new gear geometries or materials.

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